

RESEARCH ARTICLE

Muscle glycogen depletion does not alter segmental extracellular and intracellular water distribution measured using bioimpedance spectroscopy

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Shiose K, Yamada Y, Motonaga K, Takahashi H. Muscle glycogen depletion does not alter segmental extracellular and intracellular water distribution measured using bioimpedance spectroscopy. *J Appl Physiol* 124: 1420–1425, 2018. First published February 8, 2018; doi:10.1152/jappphysiol.00666.2017.—Although each gram of glycogen is well known to bind 2.7–4.0 g of water, no studies have been conducted on the effect of muscle glycogen depletion on body water distribution. We investigated changes in extracellular and intracellular water (ECW and ICW) distribution in each body segment in muscle glycogen-depletion and glycogen-recovery condition using segmental bioimpedance spectroscopy technique (BIS). Twelve male subjects consumed 7.0 g/kg body mass of indigestible (glycogen-depleted group) or digestible (glycogen-recovered group) carbohydrate for 24 h after a glycogen-depletion cycling exercise. Muscle glycogen content using ¹³C-magnetic resonance spectroscopy, blood hydration status, body composition, and ECW and ICW content of the arm, trunk, and leg using BIS were measured. Muscle glycogen content at the thigh muscles decreased immediately after exercise (glycogen-depleted group, 71.6 ± 12.1 to 25.5 ± 10.1 mmol/kg wet wt; glycogen-recovered group, 76.2 ± 16.4 to 28.1 ± 16.8 mmol/kg wet wt) and recovered in the glycogen-recovered group (72.7 ± 21.2 mmol/kg wet wt) but not in the glycogen-depleted group (33.2 ± 12.6 mmol/kg wet wt) 24 h postexercise. Fat-free mass decreased in the glycogen-depleted group ($P < 0.05$) but not in the glycogen-recovered group 24 h postexercise. However, no changes were observed in ECW and ICW content at the leg in both groups. Our results suggested that glycogen depletion per se does not alter body water distribution as estimated via BIS. This information is valuable in assessing body composition using BIS in athletes who show variable glycogen status during training and recovery.

NEW & NOTEWORTHY Segmental bioimpedance spectroscopy analysis reveals the effect of muscle glycogen depletion on body segmental water distribution in controlled conditions. Despite the significant difference in the muscle glycogen levels at the leg, no difference was observed in body resistance and the corresponding water content of the extracellular and intracellular compartments.

bioimpedance; body composition; glycogen

INTRODUCTION

Muscle glycogen is consumed as energy fuel during exercise, and its recovery is fueled by postexercise carbohydrate intake (6). Muscle glycogen levels are presumed to be changed

substantially via a large amount of training and following recovery in athletes.

Previous animal and human studies demonstrated that the increase in muscle glycogen following high-carbohydrate diet (i.e., carbohydrate loading) is accompanied by an increase of 2.7–4.0 g of body water per gram of glycogen (11, 12, 25, 26). In addition, studies using magnetic resonance imaging with T2 relaxation time and bioimpedance techniques have demonstrated that the increased body water during carbohydrate loading is mainly related to the increase in intracellular water (4, 22, 30). These results suggest that glycogen is stored with intracellular binding water. However, studies on changes in body water distribution in muscle glycogen depletion state are limited.

Segmental bioimpedance spectroscopy (BIS) is a simple technique to estimate whole body and segmental intracellular water (ICW) and extracellular water (ECW) based on body resistance (17). In accordance with accuracy, a relatively wide range of limit of agreement between gold standard methods (isotope dilution techniques and whole body ⁴⁰K count) and whole body BIS was reported in a study conducted in children and young adults (10), but it may partly improve using segmental BIS (37). For the group mean level, good accuracy of the whole body and segmental BIS on body water assessment is accepted (9, 35, 37). In addition, it was previously shown to be a sensitive method in detecting changes in segmental ICW and ECW during carbohydrate loading (30).

This study aimed to clarify the effects of glycogen depletion on segmental water distribution using BIS with glycogen recovery condition as a control. We hypothesized that glycogen depletion did not alter body water distribution because intracellular binding water was tightly structured not only glycogen but also other cytosolic macromolecules.

METHODS

Subjects. Twelve healthy men [age, 21.6 ± 1.4 yr; height, 174.7 ± 6.4 cm; body mass (BM), 65.7 ± 6.1 kg; BM index, 21.6 ± 2.3 kg/m²; and maximal oxygen uptake ($\dot{V}O_{2max}$), 50.1 ± 6.1 ml·kg⁻¹·min⁻¹] participated in this study. Written informed consent was obtained from all subjects, and the study procedure was approved by the ethical committee of the Japan Institute of Sports Sciences.

Experimental procedures. Subjects initially completed a preliminary exercise test on a bicycle ergometer (Excalibur Sports, Lode, Groningen, The Netherlands) to determine $\dot{V}O_{2max}$. The exercise test followed a 3-min warm up at 40 W, and the work load was increased by 20 W every minute until voluntary exhaustion. Expired gas was analyzed using a calibrated online gas analyzer (AE-310S; Minato

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Medical Science, Osaka, Japan). $\dot{V}O_{2\max}$ was determined according to previous described criteria (2). Thereafter, subjects completed weighted dietary records for 3 separate days, including 2 week days and 1 weekend day. A well-trained dietitian analyzed the dietary records based on standard Japanese tables of food composition using a software program (Excel Eiyo-kun, ver.8; Kenpakusha, Tokyo, Japan).

On the day preceding the experiment, subjects received verbal and e-mail instructions prohibiting vigorous exercise, excessive water intake, and consumption of any alcoholic drink. Subjects arrived at the laboratory on the eve of the experiment and stayed overnight to standardize their physical activity and wake-up time the following day.

On the day of the experiment, subjects were asked to wake up 1 h before the experiment (0700–0900) and drink 285 ml of water within 30 min after waking up. The baseline measures of body composition, muscle glycogen content, and body water content were taken along with blood samples for analysis. Thereafter, subjects completed the glycogen-depletion exercise in accordance with a previous study (30). Briefly, subjects completed 60 min of continuous exercise along with high-intensity intermittent exercise on a bicycle ergometer until exhaustion. Muscle glycogen content was measured immediately following the exercise. For the next 24 h, subjects stayed at the laboratory and were randomized to receive a prescribed diet with either indigestible carbohydrates (erythritol; glycogen-depleted group) or digestible carbohydrate (maltodextrin, glucose, and fructose; glycogen-recovered group). Designated carbohydrate (1.5 g/kg BM) was provided immediately after and 6 h after the exercise, and 2.0 g/kg BM was provided 2 and 10 h after the exercise. These carbohydrates were provided using protein- and fat-free jelly. In addition, a total of 50 g of digestible carbohydrate, which is equivalent to 0.8 ± 0.1 g/kg BM, was contained in meals provided at 2 and 10 h after exercise in both groups. Subjects could drink water ad libitum between meals. Dietary composition and water volume drunk are shown in Table 1. Measurements identical to those taken at baseline were completed 24 h postexercise. All subjects were asked to crossover to the other group after a minimum of 1 wk for a repetition of the test. Daily schedule and water intake between meals were strictly standardized between groups.

Muscle glycogen content. Muscle glycogen content was measured using ^{13}C -magnetic resonance spectroscopy technique in accordance with a previous study (30, 33). Briefly, ^{13}C -glycogen spectra from the right thigh muscles were collected on a 3-Tesla magnetic resonance system (Magnetom Verio; Siemens, Erlangen, Germany) with a ^{13}C - ^1H double-tuned surface coil of a 10-cm diameter. ^{13}C -spectra were acquired in 15-min blocks consisting of 4,500 scans and a repetition time of 200 ms. The quantification of muscle glycogen content was obtained by performing a phantom replacement measurement. The phantom was prepared with a 120 mM of glycogen and 50 mM potassium chloride (KCl), and the diameter of the phantom was adjusted to that of the thigh scanned on the magnetic resonance. The

peak area of the ^{13}C -glycogen signal was integrated by commercial software provided with the magnetic resonance console.

Body composition. BM and body density were measured using an air displacement plethysmography (ADP) system (BOD POD; COSMED, Chicago, IL). Thoracic gas volume was measured at baseline and 24 h postexercise, and the mean value of the measures was applied for the calculation of body density. Fat and fat-free mass were calculated by using the Brozek's equation (5). For four subjects, whole body total water content measurement using isotope ($^2\text{H}_2\text{O}$ and H_2^{18}O) dilution method (TBW_{IDM}) was determined in accordance with previous studies (29, 30). Briefly, subjects randomly ingested 0.12 g of $^2\text{H}_2\text{O}$ or 0.25 g of H_2^{18}O /kg of their predicted TBW. Predicted TBW was estimated based on the body weight with assumptions that percent fat mass is 15% and $\text{TBW}/\text{fat-free mass}$ (FFM) ratio is 0.732. Serum samples were collected at baseline and 3 and 4 h after ingestion. Isotope abundances in the blood were measured using an isotope-ratio mass spectrometer (Hydra 20-20 Stable Isotope Mass Spectrometer; Sercon, Crewe, UK).

BIS. Whole body and segmental water contents were measured using the BIS system (SFB7; ImpediMed; Pinkenba, QLD, Australia) in accordance with a previous study (30). Briefly, 30×20 mm electrodes (Red Dot; 3M Health Care, St. Paul, MN) were placed on the right side of the body as injection and sensing electrodes, respectively. The electrode position was marked and strictly standardized for each measurement. Following the 15-min rest in supine position in a temperature-controlled room ($21\text{--}22^\circ\text{C}$), body resistance was measured for the whole body (wrist-to-ankle) and the arm, trunk, and leg segments.

The analysis parameters were set in accordance with a previous study (36). Extracellular and intracellular resistance components (R_e and R_i , respectively) were estimated using the Cole plot (7). The whole body fat mass, FFM, and extracellular and intracellular water contents (ECW and ICW) were calculated with a manufacturer-provided equation (based on the Hanai mixture theory) (8, 15), using a commercial analysis software (ImpediMed SFB7 Multi-Frequency Analysis; ImpediMed). The specific resistance (ρ) of ECW and ICW was assumed to be 273.9 and 937.2 $\Omega\cdot\text{cm}$, respectively, and constant body density (Db) and body proportion factors (Kb) were applied for the calculation ($\text{Db} = 1.05$ kg/l and $\text{Kb} = 4.30$, respectively). ECW and ICW of the arm, trunk, and leg were calculated according to Zhu's equation with segment-specific resistivity (37). TBW was calculated as sum of ECW and ICW.

The day-to-day variation as assessed by the coefficient variance in ECW and ICW in this design was calculated for the whole body (1.7 and 0.7%), arm (2.4 and 2.2%), trunk (2.6 and 5.6%), and leg (2.9 and 1.1%).

Blood and urine analysis. Blood samples were collected from the antecubital vein into vacuum tubes containing lithium-heparin and dipotassium EDTA. The samples in the lithium-heparin tubes were centrifuged at 3,000 g at 4°C for 10 min and analyzed for plasma osmolality and sodium, potassium, and chloride concentrations. The samples in the dipotassium EDTA tubes were stored at 4°C and analyzed for hematocrit. Urine samples were stored at 4°C and used for urine-specific gravity analysis. All blood and urine analyses were conducted in an independent laboratory (LSI Medience, Tokyo, Japan).

Statistical analyses. All data were presented as means $\pm$ SD. Glycogen units were converted from millimoles per liters to millimoles per kilograms of wet weight by assuming muscle density as 1.1 kg/l (18). A repeated-measures ANOVA with two within-subject factors (time and condition) was performed. When a significant interaction was identified, data were subsequently analyzed using a Bonferroni post hoc test. The statistical significance was set at $P < 0.05$. Statistical calculations were performed using SPSS Statistics version 24 (IBM). TBW_{IDM} was excluded from the statistical analysis because of insufficient sample size ($n = 4$).

Table 1. Dietary composition and water intake

	Glycogen-Depleted Group	Glycogen-Recovered Group
Energy, kcal	1,063	2,903 $\pm$ 171*
Protein, g	79	79
Fat, g	61	61
Carbohydrate, g	510 $\pm$ 43	510 $\pm$ 43
Digestible carbohydrate, g/kg	0.8 $\pm$ 0.1	7.8 $\pm$ 0.1*
Indigestible carbohydrate, g/kg	7.0	0.0*
PFC ratio	30:51:19	11:19:70
Water intake, g	755 $\pm$ 405	756 $\pm$ 389

Values are numbers or means $\pm$ SD. PFC, protein, fat, and carbohydrate. * $P < 0.05$, between the 2 groups.

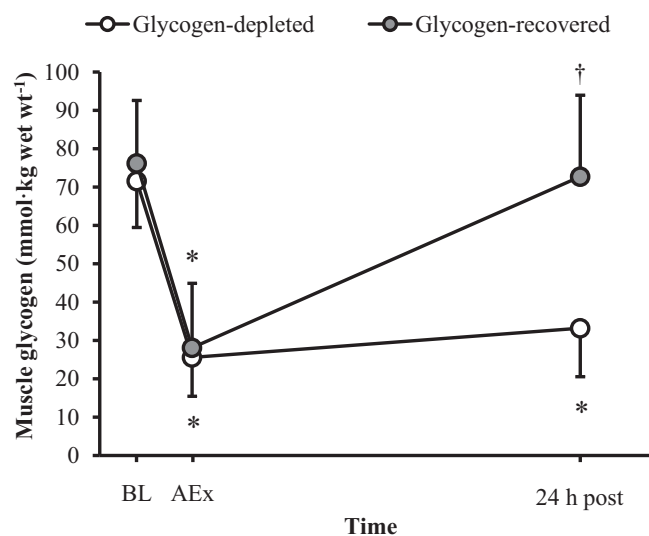


Fig. 1. Changes in muscle glycogen content. BL, baseline; AEx, immediately after exercise; 24 h post, 24 h postexercise. * $P < 0.05$, significantly different from the baseline within the same group. † $P < 0.05$, significantly different between the 2 groups.

RESULTS

Muscle glycogen. Figure 1 shows the changes in muscle glycogen content. A significant interaction was observed between the condition and time for muscle glycogen [$F(1,11) = 143.87$; $P < 0.001$; partial $\eta^2 = 0.929$]. Baseline muscle glycogen content was similar between the glycogen-depleted (71.6 ± 12.1 mmol/kg wet wt) and glycogen-recovered groups (76.2 ± 16.4 mmol/kg wet wt). This value was significantly decreased following exercise in the glycogen-depleted group (25.5 ± 10.1 mmol/kg wet wt; $P < 0.001$) and glycogen-recovered group (28.1 ± 16.8 mmol/kg wet wt; $P < 0.001$), compared with baseline. At 24 h postexercise, muscle glycogen content recovered in the glycogen-recovered group, compared with baseline (72.7 ± 21.2 mmol/kg wet wt; $P = 0.922$) but

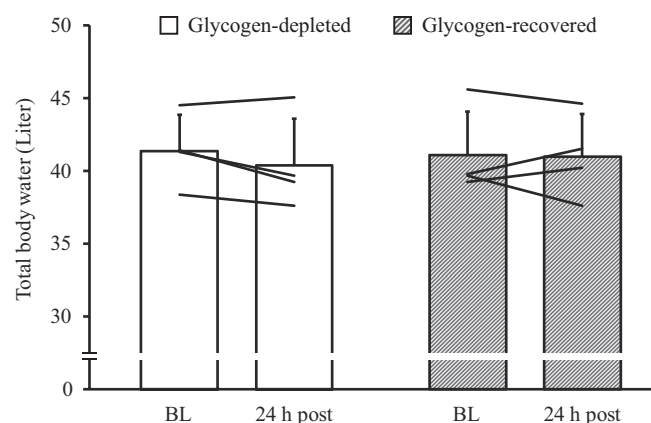


Fig. 2. Changes in body water content assessed using isotope dilution technique. BL, baseline; 24 h post, 24 h postexercise. Lines denote individual data of 4 subjects in each group.

not in the glycogen-depleted group, compared with baseline (33.2 ± 12.6 mmol/kg wet wt; $P < 0.001$).

Body composition. Table 2 shows the changes in body composition. A significant interaction was observed between the condition and time for BM [$F(1,11) = 30.88$; $P < 0.001$; partial $\eta^2 = 0.737$] and FFM as assessed via ADP [$F(1,11) = 30.88$; $P = 0.014$; partial $\eta^2 = 0.438$]. BM and FFM were decreased in the glycogen-depleted group 24 h postexercise (both $P < 0.05$) but not in the glycogen-recovered group. Changes in TBW_{IDM} are shown in Fig. 2. Baseline TBW_{IDM} was similar between the glycogen-depleted (41.4 ± 2.5 liters) and glycogen-recovered groups (41.1 ± 3.0 liters). Twenty-four hours after exercise, TBW_{IDM} was 40.4 ± 3.2 l in glycogen-depleted group and was 41.0 ± 2.9 l in glycogen-recovered group. TBW_{IDM} decreased in three out of four subjects in glycogen-depleted group and decreased in two out of four subjects in glycogen-recovered group after 24 h after exercise.

Body water content assessed using BIS. Changes in the whole body resistance and water content are shown in Table 2. No significant interaction was observed between the condition

Table 2. Changes in body composition, whole body of resistance, and water content

	Glycogen-Depleted Group		Glycogen-Recovered Group	
	Baseline	24 h postexercise	Baseline	24 h postexercise
Height, cm	174.7 ± 6.4			
Body mass,* kg	65.4 ± 5.7	64.6 ± 5.8†	65.2 ± 5.5	65.0 ± 5.4
Body density, g/ml	1.072 ± 0.009	1.073 ± 0.009	1.073 ± 0.008	1.073 ± 0.007
TBW _{IDM} , liter	41.4 ± 2.5	40.4 ± 3.2	41.1 ± 3.0	41.0 ± 2.9
%FAT _{ADP} , %	12.0 ± 3.4	11.8 ± 3.4	11.8 ± 3.2	11.6 ± 2.9
FM _{ADP} , kg	8.0 ± 2.7	7.7 ± 2.6	7.8 ± 2.4	7.6 ± 2.1
FFM _{ADP} ,* kg	57.4 ± 4.4	56.8 ± 4.6†	57.4 ± 4.1	57.4 ± 4.5
%FAT _{BIS} , %	13.6 ± 3.8	13.4 ± 4.1	12.9 ± 4.1	13.4 ± 4.1
FM _{BIS} , kg	9.0 ± 2.8	8.7 ± 2.9	8.5 ± 2.8	8.8 ± 2.9
FFM _{BIS} ,# kg	56.4 ± 4.7	55.8 ± 4.9	56.7 ± 4.6	56.1 ± 4.4
R _e ,# Ω	614 ± 56	636 ± 63	608 ± 56	616 ± 62
R _i , Ω	1,192 ± 242	1,171 ± 239	1,180 ± 238	1,190 ± 213
ECW _{BIS} ,# liter	17.4 ± 1.3	16.9 ± 1.3	17.5 ± 1.1	17.3 ± 4.6
ICW _{BIS} , liter	23.9 ± 2.5	24.0 ± 2.5	24.0 ± 2.5	23.8 ± 2.8
TBW _{BIS} , liter	41.3 ± 3.4	40.9 ± 3.6	41.5 ± 3.3	41.1 ± 4.1

Values are presented as means ± SD. %FAT, total body water measured by isotope dilution method; TBW_{IDM}, percent fat mass; FM, fat mass; FFM, fat-free mass; R_e, extracellular resistance; R_i, intracellular resistance; ECW_{BIS}, extracellular water measured by BIS; ICW_{BIS}, intracellular water measured by BIS; TBW_{BIS}, total body water measured by BIS. TBW_{IDM} was measured in 4 subjects. * $P < 0.05$, interaction between the condition and time. # $P < 0.05$, main effect for time. † $P < 0.05$, from baseline in the same group.

Table 3. *Changes in segmental resistance and water content*

	Glycogen-Depleted Group		Glycogen-Recovered Group	
	Baseline	24 h postexercise	Baseline	24 h postexercise
<i>R_e</i>				
Arm, # Ω	276 $\pm$ 32	285 $\pm$ 37	276 $\pm$ 32	280 $\pm$ 33
Trunk, * Ω	62 $\pm$ 4	64 $\pm$ 3†	61 $\pm$ 3	62 $\pm$ 3
Leg, Ω	276 $\pm$ 31	286 $\pm$ 31	271 $\pm$ 29	274 $\pm$ 33
<i>R_i</i>				
Arm, Ω	648 $\pm$ 180	627 $\pm$ 177	653 $\pm$ 167	652 $\pm$ 159
Trunk, Ω	80 $\pm$ 10	80 $\pm$ 9	79 $\pm$ 12	80 $\pm$ 10
Leg, Ω	497 $\pm$ 94	495 $\pm$ 98	489 $\pm$ 100	494 $\pm$ 92
ECW				
Arm, # liter	1.53 $\pm$ 0.14	1.48 $\pm$ 0.15	1.53 $\pm$ 0.14	1.50 $\pm$ 0.13
Trunk, * liter	6.92 $\pm$ 0.71	6.59 $\pm$ 0.60†	6.98 $\pm$ 0.64	6.87 $\pm$ 0.62
Leg, liter	4.66 $\pm$ 0.67	4.49 $\pm$ 0.53	4.73 $\pm$ 0.50	4.69 $\pm$ 0.54
ICW				
Arm, liter	1.96 $\pm$ 0.44	2.03 $\pm$ 0.43	1.94 $\pm$ 0.43	1.92 $\pm$ 0.35
Trunk, liter	8.43 $\pm$ 1.16	8.40 $\pm$ 0.82	8.67 $\pm$ 1.42	8.46 $\pm$ 1.23
Leg, liter	7.53 $\pm$ 1.24	7.58 $\pm$ 1.35	7.70 $\pm$ 1.43	7.59 $\pm$ 1.37
TBW				
Arm, liter	3.49 $\pm$ 0.53	3.50 $\pm$ 0.56	3.46 $\pm$ 0.53	3.42 $\pm$ 0.45
Trunk, liter	15.36 $\pm$ 1.67	14.99 $\pm$ 1.24	15.65 $\pm$ 1.83	15.33 $\pm$ 1.70
Leg, liter	12.19 $\pm$ 1.68	12.07 $\pm$ 1.72	12.43 $\pm$ 1.72	12.28 $\pm$ 1.75

Values are means $\pm$ SD. *R_e*, extracellular resistance; *R_i*, intracellular resistance; ECW, extracellular water; ICW, intracellular water; TBW, total body water. **P* < 0.05, interaction between the condition and time. #*P* < 0.05, main effect for time. †*P* < 0.05, from baseline in the same group.

and time of ECW [$F(1,11) = 4.713$; *P* = 0.053; partial $\eta^2 = 0.300$] and ICW [$F(1,11) = 2.049$; *P* = 0.180; partial $\eta^2 = 0.157$]. Baseline whole body ICW and ECW were similar between the glycogen-depleted and glycogen-recovered groups. ECW significantly decreased in both groups 24 h postexercise [main effect for time, $F(1,11) = 6.881$; *P* = 0.024; partial $\eta^2 = 0.385$].

Changes in segmental resistance and water content are shown in Table 3. Baseline ECW and ICW of the arm, trunk, and leg were similar between the glycogen-recovered and glycogen-depleted groups. A significant interaction was observed between the condition and time in ECW at the trunk [$F(1,11) = 6.147$; *P* = 0.031; partial $\eta^2 = 0.358$] but not at the arm [$F(1,11) = 1.489$; *P* = 0.248; partial $\eta^2 = 0.199$] and at the leg [$F(1,11) = 2.879$; *P* = 0.118; partial $\eta^2 = 0.207$]. A significant reduction in the ECW of the arm in both groups was observed, 24 h postexercise [main effect for time, $F(1,11) = 9.912$; *P* = 0.009; partial $\eta^2 = 0.474$]. Moreover, no significant interaction was found between the condition and time in ICW at the arm [$F(1,11) = 4.138$; *P* = 0.067; partial $\eta^2 = 0.273$], at the trunk [$F(1,11) = 0.361$; *P* = 0.560; partial $\eta^2 = 0.032$], and at the leg [$F(1,11) = 2.360$; *P* = 0.153;

partial $\eta^2 = 0.177$]. The ICW of the arm, trunk, and leg remained unchanged at 24 h postexercise.

Blood and urine analysis. Table 4 shows the changes in the measurement parameters. Baseline values of hematocrit, plasma osmolality; plasma sodium, potassium, and chloride concentrations; and urine specific gravity were similar between both groups. A significant interaction was found between the condition and time of plasma sodium [$F(1,11) = 8.313$; *P* = 0.015; partial $\eta^2 = 0.430$], chloride concentration [$F(1,11) = 5.454$; *P* = 0.039; partial $\eta^2 = 0.331$], and urine specific gravity [$F(1,11) = 31.863$; *P* < 0.001; partial $\eta^2 = 0.743$]. Plasma sodium concentration decreased in glycogen-depleted group (*P* = 0.001) but not in the glycogen-recovered group. Plasma chloride decreased in both groups (both *P* < 0.05), and the value at 24 h after exercise was lower in glycogen-depleted group (*P* = 0.001). Urine specific gravity increased at 24 h postexercise in the glycogen-depleted group (*P* = 0.002) but not in the glycogen-recovered group.

DISCUSSION

In this study, we investigated whole body and segmental water distribution using segmental BIS in muscle glycogen-depleted and glycogen-recovered states. Our results demonstrate that the distribution of ECW and ICW in the limbs is similar in the muscle glycogen-depleted and glycogen-recovered status induced by indigestible or digestible carbohydrate feeding after cycling exercise, respectively.

Despite the substantial difference in thigh muscle glycogen content between the glycogen-depleted and the glycogen-recovered groups, no difference was observed in ECW and ICW distribution at the leg. In case that decrease in muscle glycogen is accompanied by the loss of 2.7–4.0 g of water per gram of glycogen as is the case in carbohydrate loading (4, 25, 28), loss of water in this study is estimated to be 240–350 ml. It is comparable to 5.1–7.5% of ECW or 3.1–4.6% of ICW in the leg, from the standard muscle mass of a Japanese male adult (1), and exceeds day-to-day variability of ECW and ICW at the leg. Previous studies demonstrated that ECW as estimated using whole body BIS decreased by 3.5% in the glycogen-depleted state (4). However, they did not clarify whether glycogen depletion per se alters body water distribution since the glycogen recovered condition was not applied. Our results suggest that glycogen depletion per se does not lead to changes in body water distribution following exercise.

Macromolecules including glycogen account for 20–30% of cell mass (19). It has been suggested that each macromolecule holds a 0.6- to 1.0-nm layer of structured hydration shell around them and the thickness of shell exceeds intermacromol-

Table 4. *Changes in hematocrit, plasma osmolality, sodium, potassium, chloride, and urine specific gravity*

	Glycogen-Depleted Group		Glycogen-Recovered Group	
	Baseline	24 h Postexercise	Baseline	24 h Postexercise
Hematocrit, %	46.6 $\pm$ 2.6	46.1 $\pm$ 2.7	46.3 $\pm$ 1.6	45.0 $\pm$ 2.8
Plasma osmolality, mosmol/kgH ₂ O	287 $\pm$ 4	288 $\pm$ 3	287 $\pm$ 5	288 $\pm$ 4
Sodium, * meq/l	140 $\pm$ 1	139 $\pm$ 1†	140 $\pm$ 1	141 $\pm$ 2
Potassium, # meq/l	4.2 $\pm$ 0.3	4.0 $\pm$ 0.3	4.4 $\pm$ 0.4	4.1 $\pm$ 0.3
Chloride, * meq/l	103 $\pm$ 3	100 $\pm$ 2†	104 $\pm$ 2	102 $\pm$ 3†
Urine specific gravity*	1.018 $\pm$ 0.010	1.033 $\pm$ 0.008†	1.019 $\pm$ 0.010	1.020 $\pm$ 0.009

Values are means $\pm$ SD. **P* < 0.05, interaction between the condition and time. #*P* < 0.05, main effect for time. †*P* < 0.05, from baseline in the same group.

ecule distance in cytosol. Accordingly, most ICW is considered as the structured binding water (20). Although glycogen was depleted in this study, glycogen volumetric concentration in human cells is probably low (13). Therefore, it can be presumed that residual macromolecules have enough water retention capacity more than make up for glycogen depletion. In contrast to this result, we and other groups have commonly reported that muscle glycogen supercompensation increases body water content, and it is probably due to increase in ICW (4, 25, 30). The difference in body water dynamics between muscle glycogen-supercompensated and glycogen-depleted conditions implies that change in ICW induces exclusively when intercellular glycogen content apparently exceeds normal level.

Our body composition data showed that the FFM assessed using ADP in the glycogen-depleted group decreased by ~600 g at 24 h after exercise compared with baseline. Bone and muscle protein components of the FFM were unlikely to be changed owing to the short intervention period. The first possible component that may have depleted is the glycogen in the muscle and liver. Previous studies have found that liver glycogen, ~100 g, is almost completely depleted secondary to carbohydrate restriction (21, 34). Along with the muscle glycogen of the leg, decrease in glycogen at whole body was predicted to be ~200 g. The second possible component is water bonded to liver glycogen. As similar to muscle glycogen, liver glycogen is also suggested to be stored with some amount of water (11, 12). The third possible component is the intestinal residue and water of intestinal residue. A large intake of indigestible carbohydrate often induces laxative action (24). To prevent it, we used erythritol, which does not have a significant laxative action, in the glycogen-depleted diet. However, since the ingested volume exceeded the maximum dosage of erythritol proven to be free of laxative effect (24), loss of intestinal residue cannot be ruled out in the glycogen-depleted group. The second and third possibilities supported decrease in ECW at the trunk after glycogen depletion observed from segmental BIS. Relatively, low sensitivity of BIS was compared with ADP probably due to low resistance in the trunk owing to shape of body segment.

Changes in measurement time during the day, blood electrolyte concentration, muscle pH, and muscle inorganic phosphate and phosphocreatine concentrations are cited as factors affecting body resistance (23, 27, 31, 32). To prevent these effects, diet and physical activity pattern were strictly standardized, and BIS measurement was conducted at the same time the following day. Nonetheless, there are differences in change in plasma sodium and chloride concentration and urine specific gravity. The difference observed in urine specific gravity was probably not owing to difference in hydration status but the excreted erythritol in glycogen-depleted group (16). Changes in plasma sodium concentration alters body resistance independent to hydration status (28). However, since the change in plasma sodium concentration in this study was very small ($\Delta 1.58$ mEq/l), altered whole body resistance due to change in plasma sodium concentration is predicted to be no more than 2 Ω or 0.4% of resistance from previous equations (28).

The bioimpedance technique is widely used to assess the effects of diet and exercise on body composition. Changes in body water distribution are crucial for an accurate assessment of body composition in that the estimate equations basically

assume constant ECW/ICW and/or TBW/FFM ratios (3, 8, 37). In this study, we did not advocate the effects between muscle glycogen depletion per se on changes in body water distribution at whole body and at each body segment. These data were particularly important in the body composition assessment of athletes, wherein the changes in glycogen level during daily training and recovery are considerable. We acknowledge that a limitation of this study is that constant rho values were applied in body water calculation using BIS. In case exercise and dietary intervention alter structural peculiarity of ECW or ICW, body water calculation using constant rho value may cause some error. In addition, as similar to previous reports (14, 32), our day-to-day repeatability of body resistance using BIS is 1.1–2.9%. If there is decrease in muscle glycogen accompanied with smaller water than our repeatability, it is difficult to detect it using BIS. Further studies using robust body water assessment techniques (e.g., magnetic resonance imaging) may be beneficial to confirm the relationship between muscle glycogen depletion and body water distribution.

In conclusion, segmental BIS data from our study suggest that muscle glycogen depletion does not alter body water distribution of any body segment. Therefore, the decrease in muscle glycogen content is unlikely to impair the accuracy of body composition assessment using BIS. Our findings expand on the applicability of BIS for the accurate assessment of body composition in athletes.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

K.S., Y.Y., K.M., and H.T. conceived and designed research; K.S. and K.M. performed experiments; K.S., K.M., and H.T. analyzed data; K.S., Y.Y., K.M., and H.T. interpreted results of experiments; K.S. prepared figures; K.S. drafted manuscript; K.S., Y.Y., K.M., and H.T. edited and revised manuscript; K.S., Y.Y., K.M., and H.T. approved final version of manuscript.

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